

Monte Carlo Conformal Prediction in Hidden Markov Models

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1 Introduction

Conformal prediction constructs prediction sets with finite-sample coverage guarantees and mainly relies on exchangeability, and that means reordering the observations does not change their joint distribution [4]. Moreover, exchangeability is applicable on IID data but fails for Markov chains datasets because the next state depends on the current state, so the ordering of the sequence matters. Therefore, standard conformal prediction methods cannot be applied directly to hidden Markov models. Nettasinghe et al. [1] solve this problem by using partial exchangeability instead of full exchangeability. Their method was to partition the Markovian sequence first into specially constructed blocks, then permute the whole chain of blocks, and finally to use de Finetti's theorem for Markov chains [2] to justify this block-wise symmetry. This allows conformal prediction to retain finite-sample coverage guarantees in the HMM setting.

For a candidate future hidden-state sequence x , the exact method computes [1]:

$$q(x) = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S(\pi) \geq S(I)\}.$$

With Π being the set of admissible block permutations; $S(I)$ being the conformity score of the original arrangement; $S(\pi)$ being the conformity score under permutation π ; and $q(x)$ being the proportion of admissible permutations whose score is at least as large as the identity score. The main limitation of this method is that every admissible block arrangement must be evaluated; $|\Pi|$ can grow rapidly, and therefore exhaustive computation can become expensive.

The main idea of this project is to replace exhaustive enumeration of Π with Monte Carlo sampling from the same admissible permutation space, with the main goal to characterize the statistical-computational tradeoff between Monte Carlo accuracy and computational efficiency. The main questions studied are:

- How accurately does the Monte Carlo estimator approximate exact $q(x)$?
- How does approximation error affect prediction-set decisions?

- How quickly do Monte Carlo prediction sets approach the exact prediction set?
- What happens to empirical coverage?
- How much computation/runtime can Monte Carlo save?

2 Background: Exact HMM Conformal Method

Let $X_t \in \mathcal{X}$ be the hidden state at time t , and $Y_t \in \mathcal{Y}$ the corresponding observation. The HMM in this application is characterized by the transition matrix $P_{ij} = \Pr(X_{t+1} = j \mid X_t = i)$ and the observation matrix $B_{ij} = \Pr(Y_t = j \mid X_t = i)$. Now, for the available information, we have the fully observed calibration sequence $\{(X_t, Y_t)\}_{t=1}^T$ and future observations $Y_{T+1}, \dots, Y_{T+T_1}$. The goal, finally, is to construct a prediction set for the unknown future hidden-state sequence $x = (x_{T+1}, \dots, x_{T+T_1})$ [1].

As we mentioned earlier in the introduction, Nettasinghe et al. [1] used a weaker symmetry based on exchangeable blocks, motivated by de Finetti’s theorem for Markov chains [2]. For each candidate future state sequence x , they appended x to the calibration state sequence, combined it with the corresponding observations, formed an augmented state-observation sequence, and finally partitioned that sequence into (i, j) -blocks [1]. An (i, j) block can be permuted while preserving the relevant Markov dependence structure because of the symmetry property [1, 2]. If, let’s say, the hidden state sequence is 1-2-3-1-2-1, then the first block is defined as (1-2-3-1), and the second block is defined as (1-2-1).

Moreover, for every permutation π , compute a conformity score $S(\pi)$. The score is based on HMM filtering probabilities for the candidate future states given the future observations [1]. So, a smaller $S(\pi)$ means a better agreement between candidate states and observations; in contrast, a larger $S(\pi)$ means a poorer agreement. I denotes the identity permutation, and that means the original unpermuted block arrangement. For candidate x , compute the exact conformal quantity

$$q(x) = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S(\pi) \geq S(I)\},$$

and the prediction set

$$C_{1-\alpha} = \{x \in \mathcal{X}^{T_1} : q(x) > \alpha\},$$

so include x only when $q(x) > \alpha$ [1].

2.1 Computational Bottleneck

The principal computational cost of the exact procedure arises from evaluating the conformity score over all admissible block arrangements. For a single candidate future sequence, the filtering calculation over a prediction horizon of length T_1 has computational cost [1]

$$O(|\Pi|T_1|\mathcal{X}|^2),$$

where $|\Pi|$ is the number of admissible block arrangements and $|\mathcal{X}|$ is the number of hidden states. The $|\mathcal{X}|^2$ factor comes from the matrix-vector operation required at each HMM filtering step.

The main difficulty is therefore the growth of $|\Pi|$. If d usable (i, j) -blocks are available, the reduced permutation construction of Nettasinghe et al. [1] gives

$$|\Pi| = \frac{d!}{(d - T_1 - 1)!} = \prod_{j=0}^{T_1} (d - j).$$

For fixed T_1 and increasing d ,

$$|\Pi| = d^{T_1+1} \prod_{j=0}^{T_1} \left(1 - \frac{j}{d}\right) \sim d^{T_1+1}.$$

Thus, the dominant exact cost per candidate behaves approximately as

$$O(d^{T_1+1}T_1|\mathcal{X}|^2).$$

Although exhaustive enumeration can remain practical when the prediction horizon and number of usable blocks are small, it becomes increasingly expensive as the admissible permutation space grows. This motivates replacing the complete average over Π with a Monte Carlo estimate based on only M sampled block arrangements.

3 Monte Carlo Approximation

To reduce computational cost, we sample only M arrangements independently and uniformly from the same admissible permutation space, an approach closely related to Monte Carlo approximation of exhaustive permutation procedures [5], so

$$\Pi_1, \dots, \Pi_M \stackrel{\text{iid}}{\sim} \text{Unif}(\Pi),$$

and the indicator

$$Z(\pi) = \mathbf{1}\{S(\pi) \geq S(I)\}$$

is 1 if the permuted conformity score is at least as large as the identity score, and 0 otherwise. Moreover, the exact conformal quantity can be written as

$$q(x) = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} Z(\pi),$$

where

$$q(x) = \mathbb{E}[Z(\Pi^*)], \quad \Pi^* \sim \text{Unif}(\Pi).$$

This expectation representation motivates the Monte Carlo estimator

$$\hat{q}_M(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{S(\Pi_m) \geq S(I)\}.$$

The only difference between the exact and Monte Carlo methods is that the exact method evaluates all $|\Pi|$ arrangements, while Monte Carlo evaluates only M sampled arrangements. Moreover, conditional on the observed data, all $Z(\pi)$ values are fixed, and each $Z(\Pi_m)$ is Bernoulli with success probability $q(x)$. Therefore, the Monte Carlo estimator is unbiased:

$$\mathbb{E}[\hat{q}_M(x) \mid \text{data}] = q(x).$$

Conditional variance:

$$\text{Var}(\hat{q}_M(x) \mid \text{data}) = \frac{q(x)(1 - q(x))}{M}.$$

Therefore:

$$\text{SD}(\hat{q}_M) = O(M^{-1/2}).$$

Finally, the main statistical-computational tradeoff is that larger M gives a more accurate approximation while requiring more computation.

To get a finite- M concentration guarantee, we use Hoeffding's inequality [3]:

$$\Pr(|\hat{q}_M(x) - q(x)| > \epsilon \mid \text{data}) \leq 2 \exp(-2M\epsilon^2).$$

It means that large deviations between $\hat{q}_M(x)$ and $q(x)$ become increasingly unlikely as M grows, and the error probability decreases exponentially in M . Moreover, we define the decision margin as

$$\Delta(x) = |q(x) - \alpha|,$$

so a large $\Delta(x)$ means that a candidate is far from the threshold and is easy to classify, while a smaller $\Delta(x)$ means that a candidate is near the threshold and is harder to classify accurately. Decision disagreement requires an error of at least $\Delta(x)$, so

$$\Pr(\mathbf{1}\{\hat{q}_M(x) > \alpha\} \neq \mathbf{1}\{q(x) > \alpha\} \mid \text{data}) \leq \exp(-2M\Delta(x)^2).$$

In that way, prediction-set stability depends on both the Monte Carlo budget M and the distance of $q(x)$ from the threshold α .

Finally, to protect against Monte Carlo underestimation of $q(x)$ and to reduce the chance of incorrectly excluding a candidate, we define the corrected Monte Carlo prediction set:

$$\hat{C}_{1-\alpha, M}^{\text{corr}} = \{x : \hat{q}_M(x) > \alpha - \epsilon_M\},$$

with

$$\epsilon_M = \sqrt{\frac{\log(2/\delta)}{2M}}$$

as the Hoeffding error tolerance.

Finite- M coverage guarantee. The correction above also yields a finite- M coverage guarantee. Let X_{future} denote the true future hidden-state sequence. The exact HMM conformal procedure of Nettasinghe et al. [1] satisfies

$$\Pr(X_{\text{future}} \in C_{1-\alpha}) \geq 1 - \alpha.$$

For the Monte Carlo approximation, define

$$\epsilon_M = \sqrt{\frac{\log(2/\delta)}{2M}}.$$

Proposition. The corrected Monte Carlo prediction set satisfies

$$\Pr(X_{\text{future}} \in \widehat{C}_{1-\alpha, M}^{\text{corr}}) \geq 1 - \alpha - \delta.$$

Proof. For the true future sequence, Hoeffding’s inequality [3] gives

$$\Pr(|\hat{q}_M(X_{\text{future}}) - q(X_{\text{future}})| > \epsilon_M) \leq \delta.$$

Whenever $q(X_{\text{future}}) > \alpha$ and $|\hat{q}_M(X_{\text{future}}) - q(X_{\text{future}})| \leq \epsilon_M$, we have

$$\hat{q}_M(X_{\text{future}}) \geq q(X_{\text{future}}) - \epsilon_M > \alpha - \epsilon_M.$$

Therefore,

$$X_{\text{future}} \in \widehat{C}_{1-\alpha, M}^{\text{corr}}.$$

Using the exact conformal coverage guarantee and the Monte Carlo failure probability,

$$\Pr(X_{\text{future}} \notin \widehat{C}_{1-\alpha, M}^{\text{corr}}) \leq \alpha + \delta,$$

then

$$\Pr(X_{\text{future}} \in \widehat{C}_{1-\alpha, M}^{\text{corr}}) \geq 1 - \alpha - \delta.$$

□

4 Experiments

4.1 Synthetic Validation

We conducted a sequence of synthetic experiments to evaluate whether the Monte Carlo approximation reproduces the exact conformal procedure while reducing its computational cost. The experiments use binary HMMs for which exact block-permutation enumeration remains feasible in selected configurations, allowing the exhaustive method to serve as a benchmark. An initial baseline experiment reproduced the behavior of the exact implementation before introducing Monte Carlo approximation.

4.1.1 Monte Carlo Convergence

We first examined whether the Monte Carlo estimator $\hat{q}_M(x)$ converges to the exact conformal quantity $q(x)$ at the theoretical $M^{-1/2}$ rate. A single HMM realization was held fixed, and only the Monte Carlo block-permutation sampling was repeated. This isolates approximation error from randomness in the underlying HMM trajectory. For each Monte Carlo budget

$$M \in \{10, 25, 50, 100, 250, 500, 1000\},$$

we generated 200 independent Monte Carlo estimates for each candidate state.

Figure 1 shows that the approximation error decreases steadily as M increases. The empirical standard deviations closely follow the theoretical Bernoulli standard deviation

$$\sqrt{\frac{q(x)(1 - q(x))}{M}},$$

and the estimated log-log RMSE slopes are close to $-1/2$, consistent with the theoretical Monte Carlo rate

$$\text{RMSE}(\hat{q}_M) = O(M^{-1/2}).$$

The experiment therefore confirms that increasing the number of sampled permutations improves the accuracy of $\hat{q}_M$ at the expected rate.

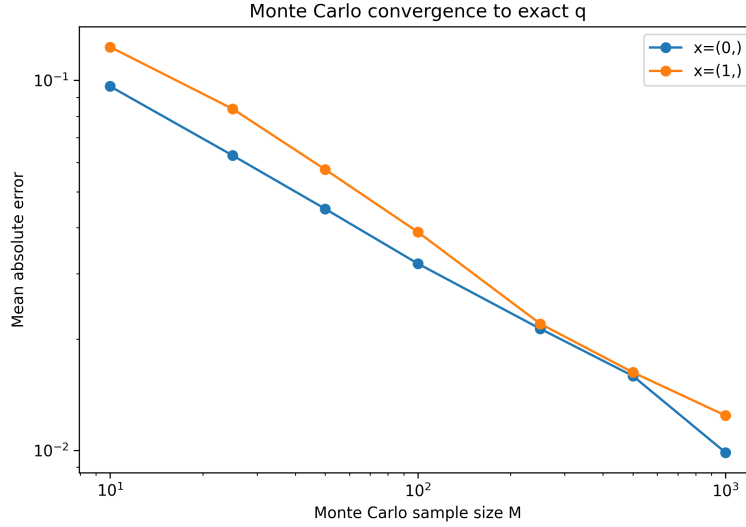


Figure 1: Monte Carlo convergence to the exact conformal quantity.

4.1.2 Prediction-Set Convergence

Accurate estimation of $q(x)$ does not necessarily imply an immediately correct prediction-set decision. Since the exact procedure includes a candidate when $q(x) > \alpha$, the difficulty of reproducing that decision depends on

$$\Delta(x) = |q(x) - \alpha|.$$

To examine this effect, we again held the HMM realization fixed and compared the exact prediction set with repeated Monte Carlo prediction sets as M increased. In the selected realization, one candidate had

$$q(x) = 0.1875, \quad \Delta(x) = 0.0125,$$

while the second had

$$q(x) \approx 0.3730, \quad \Delta(x) \approx 0.1730,$$

with $\alpha = 0.20$. The candidate far from the threshold stabilized rapidly, whereas the candidate with $q(x)$ very close to α required substantially larger M . The whole Monte Carlo prediction set eventually converged to the exact set as the sampling budget increased. This demonstrates that the practical Monte Carlo requirement is determined not only by estimation error, but also by the distance of candidate conformal scores from the inclusion threshold.

4.1.3 Computational Runtime

We next compared the runtime of exhaustive enumeration with Monte Carlo sampling. Exact computation must evaluate every relevant block arrangement, whereas Monte Carlo computation evaluates only M sampled arrangements. Consequently, the computational advantage depends on the ratio between the exact permutation count $|\Pi|$ and the chosen Monte Carlo budget.

For small permutation spaces, exhaustive enumeration can remain faster because Monte Carlo introduces sampling overhead. As the permutation space grows, however, the advantage reverses sharply. In the tested configurations, exact evaluation required approximately 4.8–5.6 seconds for permutation spaces containing roughly 5×10^4 arrangements, whereas the $M = 1000$ Monte Carlo procedure required approximately 0.10 seconds. This produced speedups of approximately 48 $\times$ and up to approximately 54 $\times$. For larger prediction horizons, the exact permutation count reached millions or billions of possible arrangements and exhaustive evaluation was skipped because of the imposed computational safety limit.

Figure 2 illustrates this transition. The results emphasize that Monte Carlo is most useful in the regime

$$M \ll |\Pi|,$$

while exhaustive enumeration may remain preferable when the admissible permutation space is already small.

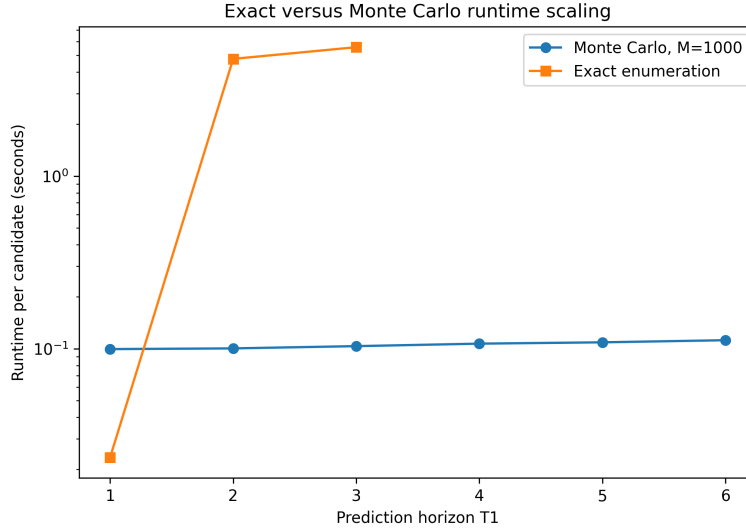


Figure 2: Runtime comparison between exact block-permutation enumeration and Monte Carlo approximation with $M = 1000$.

4.1.4 Coverage and Conservativeness

Finally, we investigated whether replacing $q(x)$ with $\hat{q}_M(x)$ materially affects empirical coverage. Three binary HMM regimes were considered: an IID/uninformative setting $(p, b) = (0.5, 0.5)$, a moderate Markov setting $(0.7, 0.75)$, and a strongly Markovian setting $(0.9, 0.75)$. We used $T = 100$, $T_1 = 1$, $\alpha = 0.20$, and compared the exact prediction set, the raw Monte Carlo set, and the Hoeffding-corrected set. The corrected procedure uses the same Monte Carlo estimates as the raw procedure and differs only through the lowered inclusion threshold.

At $M = 1000$, the empirical results were:

Table 1: Empirical coverage at $M = 1000$ for the exact, raw Monte Carlo, and Hoeffding-corrected conformal procedures.

HMM Regime	Exact	Raw MC	Corrected MC
IID / uninformative	0.870	0.863	0.900
Moderate Markov	0.820	0.827	0.853
Strong Markov	0.857	0.857	0.890

The raw Monte Carlo procedure approaches the behavior of the exact method as M increases. The Hoeffding-corrected version is systematically more conservative because lowering the threshold tends to retain additional candidate states, resulting in somewhat larger prediction sets and correspondingly higher empirical coverage. At very small M , this effect can be substantial because ϵ_M is large;

as M increases, the correction shrinks and the corrected method approaches the exact procedure.

Overall, the synthetic experiments support the theoretical picture developed in Section 3. The Monte Carlo estimator converges at approximately the $M^{-1/2}$ rate, prediction-set errors are concentrated around candidates for which $q(x)$ lies close to α , and substantial runtime reductions are possible when exhaustive block enumeration is large relative to the Monte Carlo budget.

5 Real-Data Illustration: HAPT

To test the Monte Carlo approximation method here, we apply it to the UCI HAPT smartphone dataset [6] to test whether the same behavior appears on real sequential data. The dataset contains raw accelerometer measurements and gyroscope measurements, all sampled at 50 Hz. Firstly, we aggregate the raw sensor data into non-overlapping one-second windows with 50 observations per window, and for each one-second window, we summarize the motion using the RMS change in acceleration and the RMS angular velocity from the gyroscope. Moreover, we map the original HAPT activity labels into two hidden states: either stationary, with sitting, standing, and laying, or moving, with walking, walking upstairs, walking downstairs, and postural transitions. We standardize the two sensor features using only the calibration portion of each sequence, and we discretize the resulting motion score using calibration-based quartiles. For each prediction point, $T = 150$ one-second calibration windows are used, with a one-step prediction horizon $T_1 = 1$. We then use a state-balanced evaluation design to prevent the more common state from dominating the results. For the final evaluation, we sample 32 prediction points, consisting of 16 stationary future states and 16 moving future states. Finally, as the results show, as the Monte Carlo budget increases from 25 to 1000, the MAE between Monte Carlo and exhaustive conformal quantities decreases from 0.044 to 0.008, and over the same range, the exact prediction-set agreement increases from 69% to 91%. The main conclusion is that as M increases, the Monte Carlo approximation converges toward the exhaustive procedure.

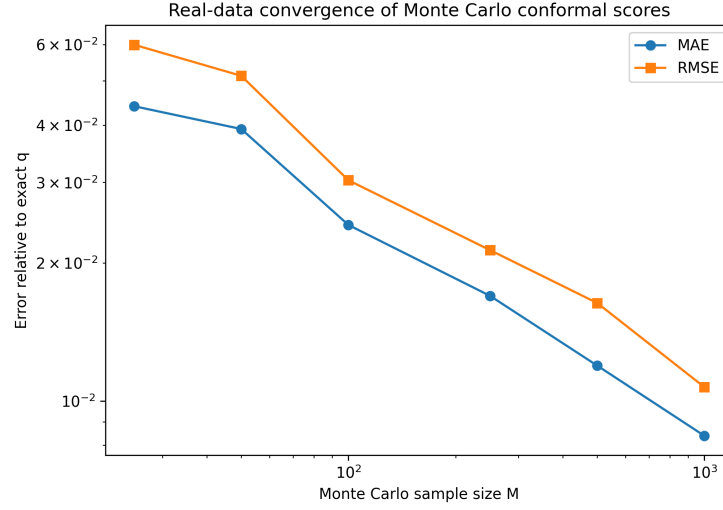


Figure 3: Convergence of Monte Carlo conformal quantities on the HAPT data.

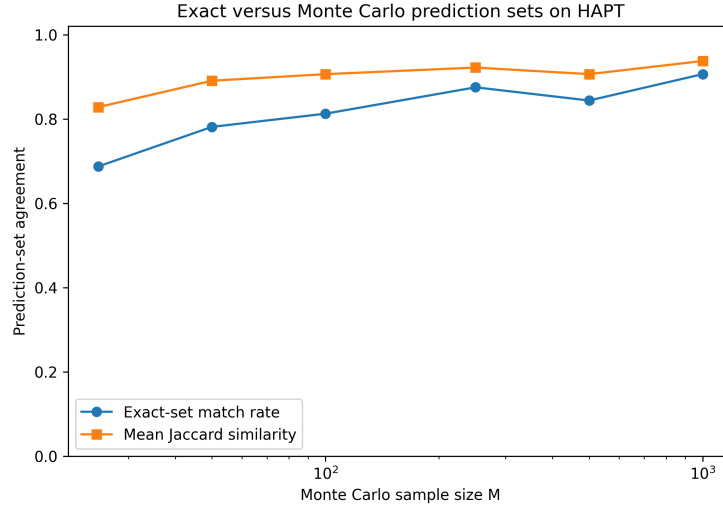


Figure 4: Agreement between Monte Carlo and exhaustive conformal prediction sets on HAPT.

6 Conclusion

In this project, we were able to show that exhaustive block-permutation conformal prediction for HMMs can be approximated using Monte Carlo sampling.

Firstly, as the number of sampled permutations M increases, the Monte Carlo estimator approaches the exact conformal quantity, and the approximation error decreases at the standard Monte Carlo rate of $O(M^{-1/2})$. Moreover, the prediction-set stability depends strongly on the decision margin $|q(x) - \alpha|$, and candidates with $q(x)$ far from α are classified reliably with relatively small M . Therefore, the greatest advantage that Monte Carlo provides in this setting is when $M \ll |\Pi|$, and that means that only a small fraction of block permutations needs to be evaluated. In permutation-heavy settings, Monte Carlo can substantially reduce runtime while maintaining close agreement with exhaustive conformal prediction. The HAPT real-data experiment shows the same statistical-computational tradeoff observed in the synthetic experiments, and that is, as M increases, the conformal-score approximation error decreases, and the Monte Carlo prediction sets increasingly agree with the exact prediction sets. Finally, the larger the M , the greater the statistical accuracy, but the greater the computation.

References

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